

Notebook 05 vs 08

MaskGIT gene programs vs Gene-Query JEPA embeddings

Same LPS atlas, same unique cells, same 1,856 HVGs. Notebook 05 = MaskGIT cls + gene varm. Notebook 08 = JEPA winner z_hat (stored as cls_embeddings) + hat varm. Overlapping analyses: neighbors(use_rep='cls_embeddings') for cells; neighbors(new_adata) + Leiden r=1 for genes. Do not treat Leiden 19 and 13 as the same cluster ID.

Shared ground

	MaskGIT 05	JEPA 08
Extract	90/10 ckpt, --split False	fzT_thenU5_vic1_contr0.3_q128_predL1 epoch=09, --eval-split all
Unique LPS cells	69,531	69,531 (10,633 / 39,185 / 19,713)
Embedding dim	768	768
Genes	1,856 HVG	1,856 HVG
Collapse	already unique in 04 dump	8.3 mean over cell_pairing_index (raw dump 444,288 rows)
Cell graph	neighbors(use_rep='cls_embeddings')	same call
Gene graph	neighbors(new_adata)	same call

05 figures: masking_split/embeddings_90_10_atlas/gene_program_discovery/figures/ and Perturbgen/docs/examples/figures/. 08 figures: gene_query_jepa/embeddings/full_vic1_contr0.3_q128_predL1_eval_all/embeddings/gene_program_discovery/figures/.

1. Load object (cls / varm)

Both: cells in obs, gene times in varm (90m_LPS, 6h_LPS, 10h_LPS), cell pool in obsm['cls_embeddings'].

08 only: cls_embeddings is mean z_hat (predicted cell pool), not an encoder CLS. z_src / z_tgt exist for extras, not for this 05 step.

Slide: same table shape, different vector (MaskGIT encoder cell vs JEPA query-pool prediction).

2. Cell UMAP (05 cell 11 <-> 08 8.3 / 8.4)

Recipe: 768-d neighbors on cls_embeddings, no PCA-50. Two 08 panels: cell type and LPS time.

What 08 shows: type dominates. CD14 monocytes are a separate island; B cells separate; T/NK sit together. Myeloid vs lymphoid is intact -- the embedding did not collapse.

Time is weaker: the shape is the type map; 90m / 6h / 10h mix inside clusters, with some 6h/10h structure in monocytes. Do not sell the time panel as 'the model learned LPS dynamics.'

Fair pair: 05 umap colored by cell_type_harmonized vs 08 first panel (umap_jepa_cell_hat_unique.png). Same question: does cell identity separate?

Not a pair: 08 src/hat/tgt stacked UMAP (PCA-50, not in 05).

3. Expression UMAP (adata_full)

Both: same FULL_H5AD, normalize_total + log1p, pca + neighbors + umap_umapcelltype.png.

This is not MaskGIT vs JEPA. It is the count atlas twice. Use it only as 'expression baseline vs embedding UMAP,' as 05's own text says.

4. Gene x time object (new_adata)

Each point = one gene at one time, 768-d mean embedding. Drop all-zero vectors: 05 = 30, 08 = 34. Then about 5.5k points. Color by timepoint; shuffle + small points so 90m does not cover 6h/10h.

Same gene at three times often lands on the same pixel if identity dominates time. That is expected in both. 08 uses hat varm only here (not src_* / tgt_*).

5. Leiden gene programs (r1, resolution 1)

	05 MaskGIT	08 JEPA hat
Clusters	31 (ids 0-30)	18 (ids 0-17)
Cytokine box	19 (139 gene x time pts)	13 (226 points)
Rest	5,399 Others	5,308 Others
Unique genes in box	48 (Immune System Enrichr)	122 in box 13

05's named box is Leiden 19, not 21. Cluster 21 in this extract is 43 genes of cell-substrate junction (UBXN11 / TSPAN2 / ...); none of the highlight cytokines sit there. Cluster 19 is Enrichr Immune System; CXCL8 at 90m plus IL32 / IL7R land there (7 highlight points). IL1B is in 4, CXCL3 in 17, IL1RN in 7, PTX3 in 2 -- the highlight list is still a drawing list, not membership. 08's named box remains 13. IDs 19 vs 13 are labels, not a quality metric.

6. Name the cytokine program

Both: typed, not automatic. 05: cluster 19 (Immune System). 08: cluster 13 (90m IL1B/CXCL8/CXCL3/IL1RN in that JEPA box).

Highlight list (IL1B, CXCL8, CXCL3, CXCL9, IL1RN, PTX3, IL32, IL7R) is a drawing list, not cluster membership.

7. Score on expression + myeloid violin

Both: score_genes on adata_full with unique symbols from the named cluster -> 'Cytokine Signaling' -> boxplot by cell type -> myeloid (CD14/CD16/DC) violin by time. Re-run 05 scoring after switching PROGRAM_CLUSTER from 21 to 19 (old violin used the wrong 43 genes).

Fair pair: 05 violinmyeloids.png vs 08 violinmyeloids.png / violin_jepa_myeloids_cytokine.png.

Aligned: 08 violin_jepa_vs_maskgit_myeloid_aligned.png -- MaskGIT Leiden 19 (48 genes) vs JEPA Leiden 13 (122 genes) on the same adata_full. Shared y. Re-score the MaskGIT side with 19. Do not read '122 > 48' as better; boxes are different sizes.

8. JEPA-only (no 05 panel)

Cell cosine vs copy (cell_cosine_vs_copy.csv):

time	n	hat-tgt	src-tgt	gap
90m	10,633	0.970	0.879	+0.091
6h	39,185	0.964	0.878	+0.086
10h	19,713	0.963	0.876	+0.087

Predictor beats copy-src at cell pool. This is the dynamics claim the time-colored UMAP cannot make.

Gene cosine (gene_cosine_vs_copy.csv): gap +0.053 to +0.061, n_shared about 1,580-1,610. Do not equate to training val/gene_gap (~+0.099) as one number.

Suggested slide order (embeddings)

1. Shared census (69,531 x 768, 1,856 genes).
2. Cell UMAP 05 vs 08 -- type separates; time weaker.
3. Gene UMAP + Leiden -- 31 vs 18 clusters; 19 vs 13 = labels.
4. Highlight -- drawing list; 05 box 19 is not every gene.
5. Myeloid aligned violin (re-scored 19 vs 13).
6. JEPA cosine table (honesty).

One line: 05 names Leiden 19 (Immune System), 08 names Leiden 13. IDs are not comparable; do not use cluster 21 (wrong box in this extract).

Decoding before KO

PerturbGen CountHead vs JEPA CountHead (notebooks 03 / 06 / 09)

Same CountHead: one cell vector -> 2k HVG ZINB counts. This is reconstruction, not gene_gap, not knockout. Show this before any IL1B KO slide.

1. Training loss (notebook 03 vs JEPA CountHead logs)

LPS_all_tps_2k, 90/10. MaskGIT count train/val uses teacher-forced true target tokens. JEPA CountHead sees predicted $\hat{z}$. Do not treat val/mse 2.7 vs 6.1 as a single crown.

	PerturbGen 03	JEPA CountHead ep39
val/loss (ZINB NLL)	1399 (best ep30)	1384
test/loss	1394	1379
val/mse	2.73	6.07
val/emd	~0.114	~0.149
gene-wise mean Pearson (test_metrics.csv)	0.991	0.987

ZINB is essentially tied (JEPA slightly lower). MSE is better for MaskGIT. Mean-profile Pearson is ~0.99 on both -- too easy to name a winner. JEPA CountHead: epoch 0 val/mse 16.5 -> epoch 39 6.07.

2. Control decode on the 90m dump (06 and 09)

CD14 monocytes, mean of per-cell Pearson (sample of 500). Notebook 09 'Control prediction sanity check' -- not the KO.

	06 MaskGIT (generate + head)	09 JEPA ($\hat{z}$ + head)
6h CD14	0.357 (n=10,424)	0.871 (n=10,402)
10h CD14	0.515 (n=7,829)	0.837 (n=7,822)

You can say: control transcriptome is closer to measured under JEPA. You cannot say: better knockout.

Suggested slide order (decoding, then KO)

1. Same CountHead; only the cell vector changes.
2. Loss table: ZINB tied; MSE with the teacher-force note.
3. 06 vs 09 control Pearson (the decoding slide).
4. Then KO as a separate question -- cell-pool, not a bake-off.

One line: JEPA CountHead matches PerturbGen on ZINB and reconstructs the 90m-source control much closer to measured counts (0.87 / 0.84 vs 0.36 / 0.52). That is decoding. It is not a knockout win, and it is not gene_gap.